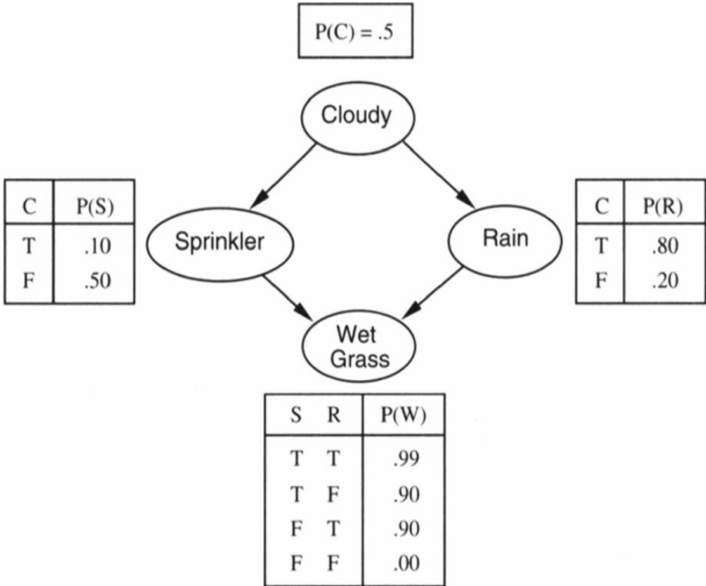


Class Test # 2
Full Marks: 20, Time: 20 minutes

ID: _____ Full Name: _____

Q1. (5 Marks) Given the Bayesian network, write *true* or *false* below/beside each of the following statements.

- I. Sprinkler and Rain are independent.
- II. Sprinter and Rain are independent given Cloudy.
- III. WetGrass is independent of Cloudy given Sprinkler and Rain.
- IV. Sprinkler is independent of WetGrass given Cloudy.
- V. Rain is independent of Sprinkler given WetGrass.



Q2. For the given Bayesian network in Q1, answer the following:

- a) (3 Marks) Write down the full join probability expression for the Bayesian network.
- b) (6 Marks) Compute the probability: $P(WG = True, R = True)$ [WG=WetGrass, R=Rain, Cloudy=C, Sprinkler=S]

Q3. (6 Marks) The following figure shows data about whether a customer will wait for a table in a restaurant or not. Compute the information gain for Price attribute at the root node.

Example	Attributes										Target	
	Alt	Bar	Fri	Hun	Pat	Price	Rain	Res	Type	Est	Will	Wait
X ₁	T	F	F	T	Some	\$\$\$	F	T	French	0-10	T	
X ₂	T	F	F	T	Full	\$	F	F	Thai	30-60	F	
X ₃	F	T	F	F	Some	\$	F	F	Burger	0-10	T	
X ₄	T	F	T	T	Full	\$	F	F	Thai	10-30	T	
X ₅	T	F	T	F	Full	\$\$\$	F	T	French	> 60	F	
X ₆	F	T	F	T	Some	\$\$	T	T	Italian	0-10	T	
X ₇	F	T	F	F	None	\$	T	F	Burger	0-10	F	
X ₈	F	F	F	T	Some	\$\$	T	T	Thai	0-10	T	
X ₉	F	T	T	F	Full	\$	T	F	Burger	> 60	F	
X ₁₀	T	T	T	T	Full	\$\$\$	F	T	Italian	10-30	F	
X ₁₁	F	F	F	F	None	\$	F	F	Thai	0-10	F	
X ₁₂	T	T	T	T	Full	\$	F	F	Burger	30-60	T	